CLOSED IMMERSION CHECKED INFINITESIMALLY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3751-3763, following the statement at 3737-3749.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Reduced and scheme criterion. lemma-reduction-scheme (same chapter, cited at ~3756-3758). X x_B U is a scheme once X' x_B U is a scheme iso on reduced spaces.
- Scheme-level infinitesimal criterion. Morphisms, Lemma morphisms-lemma-check-closed-infinitesimally (cross-chapter, cited at ~3761-3763). Concluding engine after reduction to schemes.

Source reference. Original lemma 07VU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3699–3699. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0896: focused-seeds/geometry/S9-spaces-limits-check-closed-infinitesimally/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ and $B' \to B$ be morphisms of algebraic spaces over S. Assume*

(1) *$B' \to B$ is a closed immersion,*
(2) *$|B'| \to |B|$ is bijective,*
(3) *$X \times_B B' \to B'$ is a closed immersion, and*
(4) *$X \to B$ is of finite type or $B' \to B$ is of finite presentation.*

Then $f : X \to B$ is a closed immersion.

Proof. Assumptions (1) and (2) imply that $B_{red} = B'_{red}$. Set $X' = X \times_B B'$. Then $X' \to X$ is closed immersion and $X'_{red} = X_{red}$. Let $U \to B$ be an étale morphism with U affine. Then $X' \times_B U \to X \times_B U$ is a closed immersion of algebraic spaces inducing an isomorphism on underlying reduced spaces. Since $X' \times_B U$ is a scheme (as $B' \to B$ and $X' \to B'$ are representable) so is $X \times_B U$ by Lemma 07VU. Hence $X \to B$ is representable too. Thus we reduce to the case of schemes, see Morphisms, Lemma 0896. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.